PLANRS ACADEMY

MATHS & SCIENCE OLYMPIAD CHALLENGE

Complete Strategy-Led Master Lesson Guide • Grade 3

Session 1: Foundations of Olympiad Analytics

Duration: 30 Minutes Core Video / Full Workbook Framework

CONFIDENTIAL CONTENT PRODUCTION INTERNAL DOCUMENT

Standard Operating Curriculum Framework v4.2 • Academic Year 2026-2027

1. Pedagogical Philosophy & Blueprint

The Planrs Academy Olympiad Challenge is structurally designed to move students from passive, procedural school-level learning to deep schematic processing. At the Grade 3 level (ages 7 to 9), students are developing cognitive operational structures that allow them to process multiple abstract variables simultaneously. Rather than introducing complex algebraic parameters or technical jargon that encourages rote memorization, this curriculum systematically introduces **Higher-Order Thinking Skills (HOTS)** through integrated mathematical, scientific, and deductive reasoning modules.

Olympiad examinations like the IMO, NSO, ASSET, and Unified Council do not simply test whether a student can add numbers or list the physical states of a substance. They test adaptive reasoning—the ability to identify hidden governing laws, filter out irrelevant "distractor" parameters, and apply the process of elimination under optimized time constraints.

> **Core Interdisciplinary Core Map**
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> This master document deconstructs Session 1 into highly rigorous modules that bridge natural mathematical progressions, the microscopic particulate laws of chemistry/physics, and foundational relational data mapping.

2. Cognitive Learning Objectives

Every element of content produced under this brief must structurally satisfy and explicitly track the following cognitive learning criteria mapped to the Revised Bloom's Taxonomy:

Objective ID	Bloom's Level	Measurable Performance Indicator
LO-01 (Math)	Analyzing / Applying	Deconstruct complex multi-variable geometric and spatial patterns containing at least two shifting criteria (rotation + element growth) to accurately isolate and predict the n-th term.
LO-02 (Science)	Analyzing / Evaluating	Classify unknown materials into particulate phases (Solid, Liquid, Gas) by evaluating macro-properties (compressibility, container-dependency, shape stability) linked to micro-molecular constraints.
LO-03 (Logic)	Evaluating / Creating	Synthesize multi-step logical relational constraints (ranking matrices and non-linear paths) to map absolute positional truths using systematic deduction and exclusion methods.

3. Timed Session Curricular Sequence

The core interactive framework spans a strict 30-minute timeline. Content production teams must maintain precise pacing to prevent viewer fatigue and maximize learning retention spikes.

Time Segment	Module Focus	Strategic Objective & Narrative Element
00:00 – 03:00	The Hook & The Mystery	Introduction of the *Kolam / Rangoli Anomaly* and the *Winter Ghee Conundrum*. Challenges students to become analytical detectives.
03:00 – 10:00	Math Core: Pattern Decryption	Deconstructing multi-variable geometric progressions (Shape + 90° Clockwise Rotation + Odd Number Variable Arithmetic Increments).
10:00 – 17:00	Science Core: Molecular Detectives	Micro-scale visualization of structural molecules inside a standard iron *Tawa*, a steel *Katori* of liquid oil, and dispersing *Agarbatti* gas.
17:00 – 24:00	Logical Reasoning Matrix	Multi-variable relational ranking problem mapping family seating configurations (*Pangat* arrangement constraints).
24:00 – 28:00	Guided Sprint & Live Strategy	Introduction of the "Olympiad Hack"—teaching the systematic operational elimination of option traps within a 30-second window.
28:00 – 30:00	Metacognitive Wrap-Up	Consolidation of cognitive tools used and launching the asynchronous *Olympiad Arena Challenge* dashboard.

4. Module Deep-Dive: Mathematical Analytics

Concept: Multi-Variable Spatial and Geometric Progressions

In standard primary mathematics, patterns are typically limited to single linear sequences (e.g., adding 3 to every subsequent element). Olympiad questions, however, introduce layered spatial tracking. A single visual term contains multiple synchronized properties altering along separate algorithmic paths.

Olympiad Blueprint Question

Consider a sequence where Term 1 is an upright equilateral triangle containing 1 interior dot. Term 2 is the same triangle rotated 90° clockwise, containing 3 interior dots. Term 3 is rotated another 90° clockwise, containing 5 interior dots.

The Strategy: Teach students to systematically decouple the problem into individual vector parameters:

- **Parameter 1 (Orientation):** Rotational interval is constant at $R = 90^\circ ext{ Clockwise}$. Hence, the cycle repeats every 4 terms ($360^\circ / 90^\circ = 4$). Therefore, Term 5 will have the identical base structural orientation as Term 1.
- **Parameter 2 (Arithmetic Density):** Interior elements follow an arithmetic sequence governed by the formula $D_n = 2n - 1$ (an increasing sequence of consecutive odd integers starting at 1). For Term 5, the density calculation is $2(5) - 1 = 9 ext{ dots}$.

The Indian Real-World Anchor: The Mathematics of Kolam / Rangoli

To anchor this concept firmly in concrete experiences, the lesson visualizes traditional symmetrically built Indian geometric matrices (Kolams/Rangolis). Students are shown a traditional festival grid structure where lines curve symmetrically around central anchor dots. Scriptwriters must narrate this from the perspective of tracking algorithmic symmetry: how a basic 4x4 matrix configuration expands natively into an 8x8 structural form by following constant directional expansion rules.

5. Module Deep-Dive: Science Inquiry

Concept: Particulate Structural Characteristics & Phase Transitions

Grade 3 science curriculum often treats states of matter superficially by describing external shapes. The Planrs Academy Olympiad framework introduces the **particulate nature of matter** as a foundational explanatory model. Students look past macro-appearances and evaluate intermolecular spacing, relative kinetic freedom of particles, and container-dependency metrics.

Microscopic Phase Comparison Metrics

- **Solid Phase:** Microscopic particles are locked inside an unyielding, high-density matrix. Intermolecular forces are maximized; kinetic energy is restricted to subtle baseline micro-vibrations in static spatial points. This is why an iron *Tawa* (griddle) or clay *Diya* maintains absolute shape structural retention despite mechanical load forces.
- **Liquid Phase:** Intermolecular spacing is marginally expanded, allowing molecular clusters to slide, roll, and flow past each other under gravitational influence. Volume remains mathematically constant, but macro-shape becomes dependent on container geometry. This is modeled beautifully by evaluating the seasonal phase transformation of *pure Ghee* or *Coconut oil*. In cold winter months (e.g., North Indian climates), molecular thermal energy drops below a critical threshold, locking particles into a crystalline, rigid solid matrix. As summer temperatures rise, thermal absorption breaks these bonds, transitioning the material back into a fluid state that dynamically assumes the shape of a steel *Katori* or glass storage jar.
- **Gas Phase:** Particles are structurally decoupled from one another, moving completely independently at high kinetic velocities. Intermolecular bonds are functionally zero. Gas possesses neither permanent volume nor rigid structural boundaries. This property is contextualized using the rapid molecular dispersion of a lit *Agarbatti* (incense stick). The gas molecules carry aromatic compounds, escaping confinement and spreading rapidly from a dedicated Pooja corner into the entirety of the household living space via gas diffusion pathways.

6. Module Deep-Dive: Logical Relational Logic

Concept: Multi-Constraint Matrix Matching & Positional Ordering

Logical reasoning evaluates structured analytical execution under cognitive stress. Grade 3 students are introduced to spatial and linear constraints using structural tracking devices rather than guesswork.

Relational Case Scenario

Five students—Arjun, Bhavana, Charan, Diya, and Esha—are waiting sequentially in a line for a school bus at a designated stop in Bengaluru. The following structural parameters are given:

1. Arjun is positioned strictly taller than Charan, but structurally shorter than Diya.
2. Bhavana is verified to be shorter than Charan.
3. Diya is explicitly documented to be shorter than Esha.

The Strategy: Structural Ladder Construction:

Writers must teach students to avoid trying to hold all clues mentally. Instead, they build a physical relational height ladder on scratch paper, placing higher values vertically above lower ones:

[Clue 1] → Diya > Arjun > Charan | [Clue 2] → Charan > Bhavana | [Clue 3] → Esha > Diya

Synthesizing these directional vectors yields a perfect, single-threaded linear progression order:

Esha (Tallest) > Diya > Arjun > Charan > Bhavana (Shortest)

This exact schematic structure is adapted into traditional Indian seating paradigms, such as a traditional family festival gathering eating in a sequential floor *Pangat* layout. Students apply directional relative constraints ("sitting immediately to the left of Uncle Ramesh", "two positions separated from Grandmother") to map complex multi-variable grids instantly.

7. Output Engineering & Content Benchmarks

To guarantee uniformity in pedagogical transmission, all development pipelines must rigidly adhere to the following baseline output specifications:

A. Narrative & Scripting Standards

- **AV Format Protocol:** All scripts must be structured using a strict split two-column format (Left Column: Visual & Animation Assets Directive; Right Column: Auditory Voice-Over & Core Script Delivery).
- **Language Constraints:** Clear, highly engaging, conversational Indian English. Standardized localized names must be strictly utilized (e.g., Advait, Nithya, Tenzin, Fatima, Sukhvinder) to maintain geographical equity.
- **Attention Deficit Barriers:** Continuous monologue by the presenter must never exceed a 90-second run length without a mandatory graphic overlay interaction point, an explicit conceptual challenge prompt, or an animated scene structural shift.

B. Graphic Design & Video Artistry

- **Sleek Modern UI:** Avoid overly childish cartoon motifs. The aesthetic theme must project an intellectual, modern gaming environment. Background layers should focus on deep slate tones, muted indigos, and midnight blues.
- **Aesthetic Highlighters:** Use high-intensity neon color accents exclusively for marking analytical properties (Neon Cyan for vectors and line increments, Amber for molecular boundary spaces, Magenta for logical paths).
- **Text Density Boundaries:** Kinetic textual callouts are limited to a maximum length of 7 words per individual screen instance. No heavy structural paragraphs are allowed to clutter active video frames.

8. Comprehensive Olympiad Assessment Bank

The following items constitute the rigorous 15-question diagnostic matrix generated explicitly for this session framework, divided into precise difficulty tranches:

Tranche 1: Conceptual Stability (Tier 1)

Question 01 (Math): An arithmetic coin progression follows a distinct growth rule. If Step 1 contains three ₹2 coins, Step 2 contains five ₹2 coins, and Step 3 contains seven ₹2 coins, calculate the absolute monetary value of the coins present at Step 6.

A) ₹13 B) ₹22 C) ₹26 D) ₹11

Correct: C. Solution: Sequence of coin counts is 3, 5, 7... representing odd numbers. Term n count is **2n + 1**. *For Step 6, count =* **2(6) + 1 = 13 ext{ coins}**. *Total monetary worth =* **13 imes ₹2 = ₹26**. *Trap A provides only the number of coins instead of the monetary calculation.*

Question 02 (Science): Radhika applies sustained manual compression forces to three separate sealed containers containing unknown substances X, Y, and Z. Container X displays zero volume reduction. Container Y reduces its physical volume instantly by over half under moderate pressure. Container Z maintains its exact volume layout but changes shape to match the container. Identify substance Y.

A) An iron block B) Compressed air smoke C) Liquid coconut oil D) A solid wood brick

Correct: B. Solution: High compressibility is a singular distinguishing property of the gaseous phase due to massive spaces between molecules. X is a solid, Z is a liquid, and Y must be a gas. Hence, option B is the only valid gaseous match.

Question 03 (Logic): In a school lunch queue in Delhi, Aarav is standing directly behind Kabir. Kabir is standing three places ahead of Meera. If Aarav is fourth from the absolute front of the line, what is Meera's exact numerical placement from the front?

A) 5th B) 6th C) 7th D) 3rd

Correct: B. Solution: Aarav is 4th. Since Aarav is directly behind Kabir, Kabir must be 3rd. Kabir is 3 places ahead of Meera, meaning Meera's position is ***3 + 3 = 6 ext{th}****.*

Question 04 (Math): A geometric pattern block rotates exactly ***90^\circ*** counter-clockwise at each operational step. If the block starts pointing towards the absolute North orientation, where will it point after completing Step 11?

A) North B) East C) South D) West

Correct: B. Solution: Rotations repeat in cycles of 4 terms. Divide 11 by 4, which yields a remainder of 3. Three counter-clockwise movements of ***90^\circ*** *from North maps through West (1), South (2), and terminates facing East (3).*

Question 05 (Science): Why can the aroma of a freshly prepared hot paratha travel across a large household room within seconds, while a cold paratha requires you to stand closely to perceive its scent properties?

A) Hot gas particles move structurally slower.
B) High thermal energy increases molecular kinetic velocity, accelerating gas diffusion.
C) Cold substances have completely zero molecules inside them.
D) The kitchen air blocks cold particles from shifting space paths.

Correct: B. Solution: Thermal energy directly boosts kinetic energy. Higher kinetic velocity allows gas particles to move rapidly across spaces, accelerating diffusion compared to cold particles.

Tranche 2: Advanced Application (Tier 2)

Question 06 (Math): A dual-variable pattern layout changes concurrently. The outer geometric bounding box alternates structurally between a square and a circle. Simultaneously, the count of internal triangles increases by 3 at each term step. If Term 1 is a square containing 2 triangles, evaluate the exact composition of Term 6.

A) A square containing 17 triangles B) A circle containing 17 triangles
C) A square containing 20 triangles D) A circle containing 14 triangles

Correct: B. Solution: Outer tracking parameter: Odd terms = Square; Even terms = Circle. Term 6 is an even value, hence it must be a Circle. Internal triangle growth count arithmetic parameters follow: 2 + (n - 1) imes 3*. For Term 6, this equals* 2 + (5 imes 3) = 17 ext{ triangles}*. This matches Option B.*

Question 07 (Science): Amit transfers a sealed sample of coconut oil from a freezer environment (4^\circ ext{C}) into an open warm terrace area in Chennai (38^\circ ext{C}). Which accurate microstructural changes match this observation?

A) Particles lose all kinetic energy and cease movement paths completely.
B) Molecules absorb thermal energy, breaking rigid grid locks to slide past each other freely.
C) The individual mass metrics of each particle grow significantly larger.
D) The spacing shrinks down tightly, turning the substance into an impenetrable solid block.

Correct: B. Solution: Coconut oil melts when shifted from cold to hot ambient profiles. This physical phase change breaks the rigid intermolecular bonds of the solid matrix, creating a fluid liquid layout without altering particle mass.

Question 08 (Logic): Five continuous local rail stations along a Mumbai line are spaced uniformly: Vashi, Nerul, Belapur, Kharghar, and Panvel. Belapur is directly between Nerul and Kharghar. Vashi is directly adjacent to Nerul but furthest from Panvel. Identify the exact middle station in this geographic sequence framework.

A) Nerul B) Belapur C) Kharghar D) Vashi

Correct: B. Solution: Synthesizing positional parameters confirms the ordered linear layout: Vashi - Nerul - Belapur - Kharghar - Panvel. The absolute median station position belongs cleanly to Belapur, matching Option B perfectly.

Question 09 (Math): An analytical pattern line displays values on a customized number board. The steps track as follows: 4, 9, 19, 39, 79, \dots Identify the operational algorithmic growth rule and evaluate the next sequential term value.

A) Rule: Add 5 each time; Next = 84 B) Rule: Multiply by 2 then add 1; Next = 159
C) Rule: Add consecutive multiples of 10; Next = 139 D) Rule: Multiply by 3 then subtract 3; Next = 234

Correct: B. Solution: Testing structural intervals: (4 imes 2) + 1 = 9; (9 imes 2) + 1 = 19; (19 imes 2) + 1 = 39*. The progression rule is verified as* X_{n+1} = 2X_n + 1*. Following this rule, the next term is calculated as* (79 imes 2) + 1 = 159.

Question 10 (Science): An experimenter compresses a plunger inside a transparent sealed syringe barrel full of water. Even under massive manual forces, the plunger tracking line refuses to move downward. Why does this occur?

A) Water molecules are too light to be affected by mechanical force vectors.
B) Liquid molecules are already packed closely with minimal free space, making them nearly incompressible.
C) The syringe walls absorb the water instantly.
D) Liquid matter lacks any physical molecules to squeeze together.

Correct: B. Solution: Unlike gas phase elements, liquid phase entities exhibit negligible microscopic free space paths between molecular items. This packing keeps volume parameters stable under structural pressure forces.

Tranche 3: Grandmaster Higher-Order Thinking Skills (Tier 3)

Question 11 (Blended Math/Science Deep Logic): A automated climate chamber changes its state parameters based on a mathematical pattern code. The system begins at Step 1 with a solid ice block at -10^\circ ext{C}. Every subsequent operational step raises the temperature profile by exactly 4^\circ ext{C}. We know that pure water transitions from ice to liquid at 0^\circ ext{C}, and transitions to a gas at 100^\circ ext{C}. Evaluate the physical state and temperature of the chamber at Step 30.

A) Solid Ice at 110^\circ ext{C} B) Liquid Water at 106^\circ ext{C}
C) Water Vapor Gas at 106^\circ ext{C} D) Liquid Water at 120^\circ ext{C}

Correct: C. Solution: Temperature tracking calculation uses the arithmetic progression format: T_n = T_1 + (n - 1) imes \Delta T = -10 + (30 - 1) imes 4 = -10 + (29 imes 4) = -10 + 116 = 106^\circ ext{C}. Since 106^\circ ext{C} is strictly greater than the vaporization phase change boundary threshold of water (100^\circ ext{C}), the molecules have detached into the gas phase. Option C is the only fully accurate choice.

Question 12 (Spatial Matrix Logic): A 3x3 layout matrix of decorative floor tiles contains an uncompleted spatial element cell at coordinate (row 3, column 3). Moving across any row from left to right, each tile cell shape rotates exactly 90^\circ clockwise. Moving down any column from top to bottom, each tile cell adds 1 new border ring line. If the tile at row 1, column 1 is an unshaded arrow pointing absolute North with 0 border rings, determine the characteristics of the missing tile.

A) Arrow pointing West with 2 border rings B) Arrow pointing South with 1 border ring
C) Arrow pointing South with 2 border rings D) Arrow pointing East with 3 border rings

Correct: C. Solution: Track row changes to find target cell properties: row 1 moves left-to-right from North to East to South orientation. Thus, any element in row 3 will point South. Column transformation adds border lines: row 1 has 0 rings, row 2 has 1 ring, and row 3 has 2 border rings. Combining both rules requires an arrow pointing South surrounded by 2 border rings. This matches Option C.

Question 13 (Relational Sorting Elimination): On a festival day, four family members—Grandfather, Father, Deepa, and Rahul—are seated in a straight line on floor mats. Deepa is seated immediately next to Grandfather. Father is confirmed to be sitting next to Deepa, but he is not next to Grandfather. If Rahul is seated at position 1 on the absolute left edge of the sequence, calculate who is sitting at position 4 on the absolute right edge.

A) Grandfather B) Father C) Deepa D) Information insufficient

Correct: B. Solution: Positions are 1, 2, 3, 4. Rahul is at 1. The remaining positions are 2, 3, 4. Deepa sits next to Father and Grandfather. This requires Deepa to be between them (Father - Deepa - Grandfather or Grandfather - Deepa - Father). Since Father cannot sit adjacent to Grandfather, Deepa must take position 3. Since Rahul is at 1, position 2 must be adjacent to Rahul. If Grandfather is at 2, then Deepa is at 3, and Father is at 4. This fits all constraints. Thus, position 4 is occupied by Father.

Question 14 (Advanced Step Progression Mechanics): An Olympiad problem presents a multi-variable sequence matching pattern strings: Term 1 = Z1A, Term 2 = X3C, Term 3 = V5E, Term 4 = T7G. Isolate the operational parameters and predict the composition of Term 6.

A) R9I B) P11K C) P9K D) Q11J

Correct: B. Solution: Component parameter tracking analysis: 1) Initial character tracks backwards by skipping 1 letter in the English alphabet: Z, X, V, T... (Term 5 = R, Term 6 = P). 2) Central number components track consecutive odd values: 1, 3, 5, 7... (Term 5 = 9, Term 6 = 11). 3) Terminal character parameters move forward sequentially skipping 1 letter: A, C, E, G... (Term 5 = I, Term 6 = K). Merging all three parameters gives P11K, matching Option B.

Question 15 (Micro-Structural Scientific Logic): Equal amounts of thermal energy are added to three unknown material phases. Substance A's molecular particles expand their volumetric spacing by less than 0.1%. Substance B's particles break out of their structural alignment completely, increasing total volume by over 1000% instantly. Substance C's molecular bonds loosen slightly, allowing the particles to flow over one another within a fixed total volume. Classify materials A, B, and C.

A) A = Liquid, B = Solid, C = Gas B) A = Solid, B = Gas, C = Liquid
C) A = Gas, B = Liquid, C = Solid D) A = Solid, B = Liquid, C = Gas

Correct: B. Solution: Matter phase classifications determine thermal behavior: Solids (A) display strong structural retention and minimal expansion. Gases (B) undergo massive volumetric expansion due to weak molecular bonds. Liquids (C) allow internal sliding movement while maintaining a constant volume. This matches Option B.

9. Pedagogical Red Flags & Quality Enforcement

To ensure strict adherence to the brand value of Planrs Academy, all content creators must avoid specific instructional errors. Documents or video packages flagged with the following items will be rejected during internal validation reviews:

- **No Logical Shortcuts without Derivation:** Presenting procedural "tricks" or formula shortcuts without explaining their conceptual foundation is prohibited. Tricks fail when Olympiad questions alter baseline constraints. Students must always be taught the structural logic behind shortcuts.
- **No Non-Localized Word Problems:** Avoid Western textbook examples (e.g., characters named John buying 60 apples or calculating speeds in miles per hour). All context scenarios must use authentic Indian conditions, such as purchasing kilograms of Alphonso mangoes, tracking Delhi metro rail routes, or calculating runs on a cricket scoreboard.
- **No Low-Contrast Graphic Overlays:** Visual animations must be fully legible at 480p resolution on a 5.5-inch Android smartphone display. Text overlays cannot use thin fonts or light pastel colors over bright backgrounds.

10. Success Criteria & Metrics

The efficacy of production files derived from this master brief will be verified against the following metrics:

1. **Concept-to-Context Index:** 100% of the mathematical, scientific, and logical concepts introduced must be linked to a concrete real-world visual anchor within the first 60 seconds of presentation.
2. **Cohort Performance Standard:** Post-session user testing with a sample group of Grade 3 students must show a minimum average accuracy score of 82% on Tranche 1 items and 60% on Tranche 2 items.
3. **Cognitive Delight Scale:** Post-lesson interactive feedback forms must show that over 85% of active users select "Excited/Curious to solve more" over "Confused/Tired".